



2025 Spring Math

HTOP, PDE, COMPLEX, ALGORITHM

YIXIA YU

NEW YORK UNIVERSITY



EDITED BY

YIXIA YU



Contents

1	Functions of Complex Variables	2
1.1	Lecture 1 (02-03) – Complex numbers	2
1.1.1	Introduction	2
1.2	Lecture 2 (02-05) – Complex numbers	3
1.2.1	The algebra of complex numbers	3
1.3	Recitation 1(02-07)	6
2	Honors Theory of Probability	8
2.1	Lecture 1 (02-03) – Introduction to Probability	8
2.1.1	Intro	8
2.1.2	Probability Space	9
2.2	Lecture 2 (02-05) – Algebra	10
2.2.1	algebra and σ -algebra	10
2.3	Recitation 1 (02-07) – Problem Solving (Algebra)	13
2.3.1	Basic Set Theory	13
3	Partial Differential Equations	17
3.1	Lecture 1 (02-03) – What are the PDEs?	17
3.1.1	What are the PDEs?	17
3.1.2	Classical solutions	17
3.1.3	Initial and Boundary Conditions	18
3.1.4	Linear and Nonlinear waves	18
3.2	Lecture 2 (02-05) – Transport and Traveling waves	19
3.2.1	uniform transport	19
3.2.2	Nonuniform transport	20
4	Algorithm	23
4.1	Lecture 1 (02-03) – Asymptotic bounds and big O	23
4.2	Recitation 1 (02-07)–Problem Solving	24

Chapter 1

Functions of Complex Variables

§1.1 Lecture 1 (02-03) Complex numbers

§1.1.1 Introduction

You know the sets

$$(R, +, \times) \text{ and } (Q, +, \times)$$

$Q(\sqrt{2}) = \{a + b\sqrt{2} | a, b \in Q\}$ $Q(\sqrt{2})$, they are infinite sets.

Definition 1.1.1

We say that a set F together with two binary operations, called addition $+, F \times F \rightarrow F$, and multiplication $\times, F \times F \rightarrow F$ satisfying:

- associativity $(a + b) + c = a + (b + c)$
- commutativity $a + b = b + a, ab = ba$
- existence of identities
there is a $O \in F$ such that $a + O = a$ for all $a \in F$
there is a $1 \in F$ such that $a \times 1 = a$ for all $a \in F$
- existence of inverses
for every $a \in F$ there is a $b \in F$ such that $a + b = O$
for every $a \in F$ there is a $b \in F$ such that $a \times b = 1$
- distributivity
 $a \times (b + c) = a \times b + a \times c$

Example 1

- For every prime number p , and $n \geq 0, n \in N$, there is a field $(F_{p^n}, +, \times)$ with $p^n = q$ elements (cardinality q).
- $(C, +, \times)$ is a field

Question: $(R^d, +), d \geq 2$, is there a multiplication, $\times$ such that $(R^d, +, \times)$ is a field?

- R^2 , scalar product, $x \cdot y = x_1y_1 + x_2y_2 \in R$ is not a field because $R^2 \times R^2 \rightarrow R$.
- cross product, not satisfying Definition 1 and 5.
- $d=2$, there is a good definition of multiplication, $(R^2, +, \times) \cong (C, +, \times)$.

Lemma 1.1.1

Consider the vector space $(\mathbb{R}^2, +)$
 Define the multiplication: $(a, b) \times (c, d) = (ac - bd, ad + bc)$
 Then $(\mathbb{R}^2, +, \times)$ is a field.

Proof:

- You can check that $\times$ is commutative and satisfies associativity and distributivity.
- We can check $(1, 0)$ is a multiplication identity.
- also if $(a, b) \neq (0, 0)$, then

$$(a, b) \times \left(\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2} \right) = \left(\frac{a^2 + b^2}{a^2 + b^2}, \frac{-ab + ab}{a^2 + b^2} \right) = (1, 0)$$

§1.2 Lecture 2 (02-05)Complex numbers

§1.2.1 The algebra of complex numbers

It is important to classify sets:

in Set Theory: bijection

in Topology: homeomorphism

For **Fields**: isomorphism

Definition 1.2.1

We say that fields F and F' are isomorphic if there exists a function

$$i : F \rightarrow F'$$

which is a bijection and preserve the binary operations

$$i(a + b) = i(a) + i(b)$$

and

$$i(ab) = i(a)i(b)$$

.
 i is called an isomorphism and we write $F \cong F'$

Comments:

There is a unique field up to isomorphism of order (its cardinality)

$$p^n : F_{p^n}$$

Definition 1.2.2

Given a field $(F, +, \times)$, we say that $(E, +, \times)$ is a subfield of $(F, +, \times)$ if $E \subset F$ and the addition and multiplication of E is the same addition and multiplication of F .

Example 1

$$Q \subset R$$

$$Q \subset Q(\sqrt{2}) \subset R \subset R^2$$

Definition 1.2.3

Let F be a field such that:

- R is a subfield of F
- there is an element $i \in F$ which is the root of the equation

$$x^2 + 1 = 0 (i^2 = -1)$$

Then we define the field of complex numbers C as

$$C = \{x \in F : x = \alpha + i\beta, \alpha, \beta \in R\}$$

Comments:

- We need to show that there is at least one field F satisfying these properties.
Actually, $(F, +, \times) = (R, +, \times)$ satisfies these properties.
- $R' = \{x \in R : x = (\alpha, 0), \alpha \in R\} \cong R \rightarrow R$ is a subfield of R^2 .
- Define $i := (0, 1) \in R^2$

$$(a, b) \times (c, d) := (ac - bd, ad + bc)$$

Then,

$$i^2 = (0, 1)^2 = (0, 1) \times (0, 1) = (-1, 0)$$

Example 2

Set of 2×2 real matrices of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}, a, b \in R$$

This set together with the standard addition and multiplication of matrices is a field and it is isomorphic to $(R^2, +, \times)$.

$$F = R^2, C = R^2$$

Comments:

It turns out that any definition of C is unique (up to isomorphisms).

Lemma 1.2.1

Let F and F' be two fields satisfying the properties of the definition of complex numbers, then call $\mathbb{C}$ and $\mathbb{C}'$ the corresponding complex number fields. Then $\mathbb{C} \cong \mathbb{C}'$.

Proof:

Call $i \in \mathbb{C}$ and $i' \in \mathbb{C}'$ the square roots -1 .

Then:

$f : \mathbb{C} \rightarrow \mathbb{C}'$ defined as $f(\alpha + i\beta) = \alpha + i'\beta$ is an isomorphism

$$z = \alpha + i\beta, z' = \alpha' + i'\beta', w = \gamma + i\delta, w' = \gamma' + i'\delta'$$

Lemma 1.2.2

Consider the vector space $(\mathbb{R}, +)$. Then any multiplication ' $\times$ ' defined in this vector space which transforms it into a field $(\mathbb{R}, +, \times)$ is such that $(\mathbb{R}, +, \times) \cong \mathbb{C}$.

Proof:

Consider the elements

$$(1, 0) \text{ and } j = (0, 1) \in \mathbb{R}^2$$

Then any element of $\mathbb{R}^2$ can be written as $\alpha + j\beta, \alpha, \beta \in \mathbb{R}$

Then

$$j^2 = a + jb, a, b \in \mathbb{R}$$

Then,

$$\begin{aligned} j^2 - jb = a, j^2 - 2\frac{jb}{2} = a &\rightarrow j^2 - 2\frac{jb}{2} + \frac{b^2}{4} = a + \frac{b^2}{4} \\ \left(j - \frac{b}{2}\right)^2 &= a + \frac{b^2}{4} \end{aligned} \tag{1.1}$$

Define: $i = \frac{j - \frac{b}{2}}{\sqrt{a + \frac{b^2}{4}}}$

Claim:

$$a + \frac{b^2}{4} < 0$$

If:

$$a + \frac{b^2}{4} \geq 0$$

Then (1.1) can be written as

$$\rightarrow \left(j - \frac{b}{2}\right) + \sqrt{a + \frac{b^2}{4}} \left(j - \frac{b}{2}\right) - \sqrt{a + \frac{b^2}{4}} = 0$$

either

$$j - \frac{b}{2} = \sqrt{a + \frac{b^2}{4}}, j - \frac{b}{2} = -\sqrt{a + \frac{b^2}{4}}$$

§1.3 Complex numbers

The field of rationals $\mathcal{Q}$ is the fractional field of the integral domain $\mathcal{Z}$.
With the metric

$$d(x, y) = |x - y|$$

$$|x| = \max\{x, -x\}, \forall x, y \in \mathcal{R}$$

The field $\mathcal{Q}$ is not a complete metric space.
Its completion $\mathbb{R}$, nevertheless is not algebraically closed.
For instance, $f(x) = x^2 + 1$, doesn't split in $\mathbb{R}$.
So we define the splitting field for $f(x)$ over $\mathbb{R}$
i.e. the smallest field extension of $\mathbb{R}$ such that $f(x)$ has roots.

Consider the principal ideal $\langle f(x) \rangle$ in $\mathbb{R}[x]$
since $f(x)$ is irreducible, then $\langle f(x) \rangle$ is a maximal ideal.
Then the quotient ring $\mathbb{C} = \frac{\mathbb{R}[x]}{\langle f(x) \rangle}$ is a field!
which is the splitting field of $f(x)$
The field $\mathbb{C}$ has no further splitting extensions! All polynomials in $\mathbb{C}[x]$ split in $\mathbb{C}$.
So as soon as we forced one polynomial. $f(x) = x^2 + 1$ to have roots, we in fact imposed that all polynomials have roots.

Topology:

The topology of $\mathbb{C}$ is isomorphic to $\mathbb{R}^2$.
Thus $\mathbb{C}$ is locally compact, but not compact space.
The Alexandroff compactification says that we can add a point at infinity to $\mathbb{C}$ so that $\mathbb{C} \cup \{\infty\}$ compact.

Definition 1.3.1

Alexandroff compactification

- A set $U \supseteq \mathbb{C} \cup \{\infty\}$ is an open neighborhood of ∞ if $U = \mathbb{C} \cup \{\infty\} \setminus K$ When $K \supseteq \mathbb{C}$.

$\mathbb{C} \cup \{\infty\}$ is called the Riemann sphere.

Definition 1.3.2

Holomorphic Functions

A function $f : \Omega \rightarrow \mathbb{C}$ is holomorphic where $\Omega \supseteq \mathbb{C}$ is holomorphic iff:

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) \text{ exists, } \forall z_0 \in \Omega$$

$$f(z) = f(x, y) = u(x, y) + iv(x, y), \text{ where } z = x + iy \in \Omega$$

$$u(z) = \operatorname{Re}(f(z)), v(z) = \operatorname{Im}(f(z))$$

$f : \Omega \rightarrow \mathbb{C}$ being holomorphic of course implies that $u, v : \Omega \rightarrow \mathbb{R}$ differentiable.
If we take $f(z_0) = z_0 = 0$ and we write

$$\frac{\partial}{\partial z} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right), \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$$

Then,

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{f(z)}{z} = \left(\frac{\partial f}{\partial z}\right)_{(z_0=0)} + \frac{\bar{z}}{z} \left(\frac{\partial f}{\partial \bar{z}}\right)_{(z_0=0)} + o(z)$$

For real $z \rightarrow 0$, we have $\frac{\bar{z}}{z} = 1$, and for purely imaginary $z \rightarrow 0$, we have $\frac{\bar{z}}{z} = -1$.
we have to restrict $\frac{\partial f}{\partial \bar{z}} = 0$

Cauchy-Riemann equations:

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

$$f'(z) = \frac{\partial f}{\partial z} = \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}\right)(x, y), \text{ where } z = x + iy \in \Omega$$

Nevertheless, f being holomorphic is not equivalent to say f satisfies the Cauchy-Riemann equations.

Indeed we take $f(z) = \frac{z^5}{|z|^4}$ on $\mathbb{C} \setminus \{0\}$ and $f(0) = 0$.

We can observe that $f(z)$ satisfies the Cauchy-Riemann equations on $\mathbb{C}$ but f is not even continuous at $z = 0$.

Looman Menchoff Theorem:

Theorem 1.3.1

A continuous function $f : \Omega \cup \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic iff f satisfies the Cauchy-Riemann equations.

$$\forall f, g \in H(\Omega)$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$(g \circ f)'(z) = g'(f(z)) \cdot f'(z)$$

Chapter 2

Honors Theory of Probability

§2.1 Lecture 1 (02-03) Introduction to Probability

Office Hour: Wed 12:30-1:30pm, Fri 3:30-4:30pm W910

- Homework: weekly
- Grades: 5% participation, 15% homework, 40% midterm, final group project (Max 4 people a group, presentation 20%, Final report 20% 10 pages)

§2.1.1 Intro

Why do we need modern theory of probability?

Example 1

- Coin flip 7 times, what is $P[\text{first outcome and fifth outcome are Head}]$?
sol:

$$\Omega = (x_1, x_2, x_3, x_4, x_5, x_6, x_7)$$

$$A = (x_1, x_2, x_3, x_4, x_5, x_6, x_7), x_1 = x_5 = H$$

$$P(A) = \frac{|\Omega|}{|A|} = \frac{1}{4}$$

- Stock Price (mathematically, geometric Brownian motion, Stochastic process, such that $t \rightarrow S_t$ is continuous but nowhere differentiable)
What is $P[S_T \geq 100]$?

$$\Omega = C[0, T] = \{\text{continuous function}, f[0, T] \rightarrow R\}$$

$$A = \{f \in C[0, T] | f(t) \geq 100\}$$

requires measure theory that defines $P[S_T \geq 100]$

In modern Prob, Probability Space: (Ω, F, P) , Ω is sample space, F is σ -algebra (meaningful subset of σ), P is probability measure ($P : F \rightarrow [0, 1]$).

Topics covered:

- Probability space, σ -algebra, measure, Conditional Probability and Independence
- Random variables (measurable functions), distribution
- Expectation (Lebesgue integral), Conditional distribution and expectation, functions of random variables, Radon-Nikodym Derivative
- Random walks

- generating functions, characteristic functions
- Branching processes
- Convergence of random variables, Law of large numbers, Monte-Carlo Method
- Central Limit Theorem
- Time permitting: Large deviations, Markov Chains

§2.1.2 Probability Space

(Ω, F, P)

Example 2: Coinflip and Stock Price

- Coin flip infinite times,

$$\Omega = \{(x_1, x_2, \dots), x_i = H, T\}$$

Let A be the event of 10^6 consecutive Tails,

$$A = \bigcup_{i=1}^{+\infty} \{x_i = x_{i+1} = \dots = x_{i+10^6-1} = 0, (x_i \in \Omega)\}$$

$$P[A] = 1$$

- Discrete Stock Price model, $t=0,1,2,\dots,T$
 T is maturing time, time step $\Delta t \ll T$

$$N = \frac{T}{\Delta t}$$

$$\text{price go} \begin{cases} \nearrow \text{ by factor : } e^{\sigma\sqrt{\Delta t}} \\ \searrow \text{ by factor : } e^{-\sigma\sqrt{\Delta t}} \end{cases}$$

$$\Omega = \{(x_1, x_2, \dots, x_N), x_i = 0 \text{ or } 1\}$$

Stock price at time t : $\forall \omega \in \Omega$

$$S_N(\omega) = S_0 e^{\sum_{i=1}^N x_i \sigma \sqrt{\Delta t} e^{(N - \sum_{i=1}^N x_i)(-\sigma\sqrt{\Delta t})}}$$

$$S : \Omega \rightarrow R(\text{Random Variable})$$

Event: return at T is positive but not more than 10%:

$$\{\omega \in \Omega : S_N(\omega) \in (S_0, 1.1S_0]\}$$

Example 3: Gambling

- Gambling:

- start with $\{0,1,2,\dots\}$ each time bet an integer amount
 - if amount of money =0, stays at 0
- wealth process: $\Omega = \{0,1,2,\dots\} \times \{0,1,2,\dots\} \times \dots$
wealth after time n: (random variable) $X_n : \Omega \rightarrow N, (x_1, x_2, \dots, x_n) \rightarrow x_n$
 (X_n) is a Markov Chain (future only depends on present state, but not the past)

$$\text{Event: } \{\text{State } j \text{ is reached from state } i\} \quad (2.1)$$

$$= \{\omega \in \Omega : \exists n \in N, X_0(\omega) = i, X_n(\omega) = j\} \quad (2.2)$$

$$= \bigcup_{m=1}^{+\infty} \{\omega \in \Omega : X_0(\omega) = i, X_n(\omega) = j\} \quad (2.3)$$

§2.2 Lecture 2 (02-05) – Algebra

§2.2.1 algebra and σ -algebra

in practice, want F to be closed under $\bigcap, \bigcup, ^c$

Definition 2.2.1

Let $\mathcal{A}$ be a collection of subsets of Ω , $\mathcal{A}$ is an algebra iff:

- $\Omega \in \mathcal{A}$
- if $A, B \in \mathcal{A}$ then $A \cap B \in \mathcal{A}$
- if $A \in \mathcal{A}$ then $A^c \in \mathcal{A}$

Remark 0

if $A, B \in \mathcal{A} \rightarrow A \cap B \in \mathcal{A}$, because $A \cap B = (A^c \cup B^c)^c$

Fact:

- ① $P(\Omega)$ (powerset) = $\{A : A \subset \Omega\}$ is an algebra
- ② smallest algebra/trivial algebra: $\{\emptyset, \Omega\}$
- ③ Let A_1, A_2 be two algebras of Ω

$$A_1 \cap A_2 = \{B \in \Omega : B \in A_1 \text{ and } B \in A_2\}$$

if $(A_j)_{j \in J}$ is a family of algebras, then $\bigcap_{j \in J} A_j$ is an algebra

- ④ Let $\mathcal{E}$ be any collection of subsets of Ω

$$a(\mathcal{E}) = \bigcap_{\substack{A \supseteq \mathcal{E} \\ A \text{ is an algebra}}} A$$

$a(\mathcal{E})$ is an algebra by ③

in fact, $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$

It is called the algebra generated by $\mathcal{E}$

- ⑤ Let $A \subseteq \Omega, \mathcal{E} = \{A\}$
Then

$$a(\mathcal{E}) = \underbrace{\{A, A^c, \Omega, \emptyset\}}_f$$

Proof:

- $a(\mathcal{E}) \subseteq f$
notice that f is an algebra since $f \supseteq \mathcal{E}$, therefore $f \supseteq a(\mathcal{E})$ because $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$.
- $f \subseteq a(\mathcal{E})$:

$$A \in a(\mathcal{E}), A^c \in a(\mathcal{E})$$

because $a(\mathcal{E})$ is an algebra, $\emptyset, \Omega \in a(\mathcal{E})$

- ⑥ $\pi = \{A_1, A_2, \dots, A_n\}, \Omega = \bigcup_{i=1}^n A_i, A_i \cap A_j = \emptyset$ Then:

$$a(\pi) = \left\{ \bigcup_{i \in I} A_i, \text{ for } I \subset 1, 2, \dots, n \right\} = \text{finite disjoint union of } (A_i)_{i=1}^n$$

- ⑦ Let $\mathcal{A}$ be an algebra of $\mathbb{R}$
Let $X : \Omega \rightarrow \mathbb{R}$ be a function
Then

$$\underbrace{\{X^{-1}(A), A \in \mathcal{A}\}}_{\{\omega \in \Omega : X(\omega) \in A, \text{ for some } A \in \mathcal{A}\}} \text{ is an algebra of } \Omega$$

Hint:

$$X^{-1}(A \cup B) = (X^{-1}(A)) \cup (X^{-1}(B))$$

- ⑧ $\Omega = \mathbb{R}, \mathcal{E} = \{\text{left open right closed intervals}\} = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty), & -\infty \leq a \end{cases}$

Then:

$$\begin{aligned} a(\mathcal{E}) &= \text{"finite disjoint union of elements in } \mathcal{E}\text{"} \\ &= \underbrace{\{I_1 \cup \dots \cup I_k, I_j \in \mathcal{E}, I_j \cap I_j = \emptyset(f)\}}_f \end{aligned}$$

Hint:

- $f \subseteq a(\mathcal{E})$ straightforward
- $a(\mathbb{R}) \subseteq f$
 - check f is an algebra
 - since $\mathcal{E} \subset f$, $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$, $a(\mathcal{E}) \subseteq f$
 - $(a, b]^c = \underbrace{(-\infty, a] \cup (b, +\infty)}_{\in f}$

Lemma 2.2.1

In Probability, if $(A_k)_{k=1}^n$ are disjoint events, we have $P(\bigcup_{k=1}^{+\infty} A_k) = \sum_{k=1}^{+\infty} P(A_k)$

Definition 2.2.2: σ – algebra

A σ – algebra (σ – field) $\mathcal{A}$ is an collection of subsets of Ω such that

- $\Omega \in \mathcal{A}$
- if $A_1, A_2, \dots \in \mathcal{A}$ then $\bigcup_{i=1}^{+\infty} A_i \in \mathcal{A}$
- if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$

Remark 0

if $A_1, A_2, \dots \in \mathcal{A}$ then $\bigcap_{i=1}^{+\infty} A_i \in \mathcal{A}$

σ -algebra represents the collection of information determined by partial derivatives

Example 1

- coinflips infinity many times

$$\Omega = \{(x_1, x_2, \dots) | x_i = 0, 1\} = (0, 1)^\infty$$

After observing first outcome, $\mathcal{F}_1 = \{\emptyset, \Omega, A_0, A_1\}$

Where $A_0 = \{(0, x_2, x_3, \dots), x_i = 0, 1\}$, $A_1 = \{(1, x_2, x_3, \dots), x_i = 0, 1\}$

After observing second outcome, $\mathcal{F}_2 = \{\emptyset, \Omega, A_0, A_1, A_00, A_01, A_10, A_11\}$

Where $A_{00} = \{(0, 0, x_3, x_4, \dots), x_i = 0, 1\}$, $A_{01} = \{(0, 1, x_3, x_4, \dots), x_i = 0, 1\} \dots$

After observing first n outcomes,

$$\mathcal{F}_n = \{\emptyset, \Omega, (A_j)_{j \in (\sigma 1)^n} \text{ and finite disjoint unions}\}$$

and

$$\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \subseteq \mathcal{F}_n \subseteq \dots$$

Proposition 2.2.3

- ① intersection of σ -algebras is a σ -algebra
- ② Let $\mathcal{E}$ be a collection of subsets of Ω

$$\sigma(\mathcal{E}) = \bigcap_{\substack{f \supseteq \mathcal{E} \\ f \text{ is an algebra}}} f \text{ is a smallest } \sigma\text{-algebra that contains } \mathcal{E}$$

- ③ For any collection $\mathcal{E}$, we have $\sigma(\mathcal{E}) \subseteq \sigma(\mathcal{E})$

- ④ For any collection $\mathcal{E}$, we have $\sigma(a(\mathcal{E})) = \sigma(\mathcal{E})$

Hint:

$$a(\mathcal{E}) \subseteq \sigma(\mathcal{E}) \rightarrow \sigma(a(\mathcal{E})) \subseteq \sigma(\mathcal{E})$$

$$\mathcal{E} \subseteq \sigma(a(\mathcal{E})) \rightarrow \sigma(\mathcal{E}) \subseteq \sigma(a(\mathcal{E}))$$

§2.3 Recitation 1 (02-07) – Problem Solving

§2.3.1 Basic Set Theory

Definition 2.3.1: $\cup, \cap, ^c$

De Morgan's Law:

$$\left(\bigcup_{j \in J} A_j\right)^c = \bigcap_{j \in J} A_j^c; \left(\bigcap_{j \in J} A_j\right)^c = \bigcup_{j \in J} A_j^c$$

Definition 2.3.2: $\setminus$

$$A \setminus B = A \cap B^c$$

$$\text{Then } A \setminus B = A \cap B^c = A \setminus (A \cap B) = B^c \setminus A^c$$

Remark 0

$$\bigcap_{n \geq 1} A_n = A_1 \setminus \left(\bigcup_{j \geq 2} (A_1 - A_j)\right) \text{ (ex)}$$

Definition 2.3.3: Limit Sets

Let $(A_n)_{n \geq 1}$ be a sequence of sets, then

$$B_k = \bigcup_{n \geq k} A_n, C_k = \bigcap_{n \geq k} A_n$$

Then B_k is increasing, C_k is decreasing

Define:

$$\limsup_{n \rightarrow +\infty} A_n = \lim_{k \rightarrow +\infty} B_k = \bigcap_{k \geq 1} \bigcup_{n \geq k} A_n$$

$$\liminf_{n \rightarrow +\infty} A_n = \lim_{k \rightarrow +\infty} C_k = \bigcup_{k \geq 1} \bigcap_{n \geq k} A_n$$

Definition 2.3.4: liminf, limsup of sequence

$$\limsup a_n = \inf_{k \geq 1} \sup_{n \geq k} a_n, \liminf a_n = \sup_{k \geq 1} \inf_{n \geq k} a_n$$

When $\limsup A_n = \liminf A_n$, Then we say $\lim_{n \rightarrow +\infty} A_n$ exist and $\lim_{n \rightarrow +\infty} A_n = \limsup A_n = \liminf A_n$

In probability,

$$\begin{aligned} \limsup A_n &= \{A_n \text{ occurs infinitely often}\} \\ &= \{A_n.i.o\} \\ x \in \limsup A_n &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k \text{ such that } x \in A_n \\ &\Leftrightarrow A_n.i.o \\ \liminf A_n &= \{A_n \text{ occurs eventually}\} \\ &\Leftrightarrow \exists k \in \mathbb{N}, \forall n \geq k, x \in A_n \end{aligned}$$

Remark 0

1. if A_n increases, then $\limsup A_n = \lim_{n \rightarrow \infty} A_n = \bigcup_{n \geq 1} A_n$
if A_n decreases, then $\liminf A_n = \lim_{n \rightarrow \infty} A_n = \bigcap_{n \geq 1} A_n$

2.

$$(\limsup A_n)^c = \liminf A_n, (\liminf A_n)^c = \limsup A_n$$

Exercise:

1. $A_k = \begin{cases} E, & \text{if } k \text{ is odd} \\ F, & \text{if } k \text{ is even} \end{cases}$

Then $\limsup A_n = E \cup F, \liminf A_n = E \cap F$

2. Let $f_n : \mathbb{R} \rightarrow \mathbb{R}$, Let $A = \{x \in \mathbb{R} : \lim_{n \rightarrow \infty} f_n(x) = f(x)\}$ Then,

$$\begin{aligned} A^c &= \bigcup_{k=1}^{+\infty} \bigcap_{m=1}^{+\infty} \bigcup_{n=m}^{+\infty} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\} \\ &= \bigcup_{k=1}^{+\infty} [\limsup_n \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}] \end{aligned}$$

3. Suppose that $\lim_{n \rightarrow \infty} f_n(x) = f(x), \forall x \in \mathbb{R}$
Then

$$\begin{aligned} \{x : f(x) \leq t\} &= \bigcap_{k=1}^{+\infty} \bigcup_{m=1}^{+\infty} \bigcup_{n=m}^{+\infty} \{x \in \mathbb{R} : f_n(x) < t + \frac{1}{k}\} \\ &= \bigcap_{k=1}^{+\infty} [\liminf_n \{x : f_n(x) < t + \frac{1}{k}\}] \end{aligned}$$

Proof:

1. ex2: Want

$$A = \bigcup_{k \geq 1} \bigcap_{n \geq 1} \bigcup_{m \geq n} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}$$

$$f_n(x) \rightarrow f(x) \text{ iff } \forall \epsilon > 0, \exists N \geq N, |f_n(x) - f(x)| < \epsilon$$

$$\{x : f_n(x) \rightarrow f(x)\} = \bigcap_{\epsilon > 0} \bigcup_N \bigcap_{n \geq N} \{x : |f_n(x) - f(x)| < \epsilon\},$$

USE that $\{|f_n - f| < \epsilon\}$ is monotone increasing in n

Review on mapping:

$$f : X \rightarrow Y, f^{-1} : Y \rightarrow X$$

Basic properties:

- $f(\bigcup_{i \in I} A_i) = \bigcup_{i \in I} f(A_i)$
- $f(\bigcap_{i \in I} A_i) = \bigcap_{i \in I} f(A_i)$

For (B_i) subset of Y :

- if $B_1 \subset B_2, f^{-1}(B_1) \subset f^{-1}(B_2)$
- $f^{-1}(\bigcup_{i \in I} B_i) = \bigcup_{i \in I} f^{-1}(B_i)$
- $f^{-1}(B^c) = (f^{-1}(B))^c$

Definition 2.3.5: Indicator Mapping

$$1_A : X \rightarrow \{0, 1\}$$

$$1_A(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

Exercise:

1. $1_{\limsup A_n} = \limsup 1_{A_n}$
2. $1_{\liminf A_n} = \liminf 1_{A_n}$

Proof: if

$$\begin{aligned} 1_{\limsup A_n}(x) = 1 &\Leftrightarrow x \in \limsup A_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, x \in A_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, 1_{A_n}(x) = 1 \\ &\Leftrightarrow \limsup 1_{A_n}(x) \geq 1 \text{ (Definition of limsup)} \\ &\Leftrightarrow 1_{\limsup A_n}(x) = 1 \end{aligned}$$

Exercise: Let $\mathcal{A}_1, \mathcal{A}_2$ be algebras of Ω

1. show that $\underbrace{\mathcal{A}_1 \cap \mathcal{A}_2}_{B \subset \Omega: B \in \mathcal{A}_1, B \in \mathcal{A}_2}$ is an algebra

2. show that $\underbrace{\mathcal{A}_1 \cup \mathcal{A}_2}_{B \subset \Omega: B \in \mathcal{A}_1 \text{ or } B \in \mathcal{A}_2}$ is an algebra iff $\mathcal{A}_1 \subseteq \mathcal{A}_2$ or $\mathcal{A}_2 \subseteq \mathcal{A}_1$

Proof: Suppose by contradiction that

$$\exists A_1 \in \mathcal{A}_1 \text{ but } A_1 \notin \mathcal{A}_2 \text{ and } A_2 \in \mathcal{A}_2 \text{ but } A_2 \notin \mathcal{A}_1 \text{ and } \mathcal{A}_1 \cup \mathcal{A}_2 \text{ is an algebra}$$

Therefore:

$$A_1 \cup A_2 \in \mathcal{A}_1 \cup \mathcal{A}_2, A_1 \setminus A_2 = A_1 \cap \underbrace{A_2^c}_{\mathcal{A}_1 \cup \mathcal{A}_2} \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$$A_2 \setminus A_1 = A_2 \cap A_1^c \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$\Rightarrow$ at least two of

$$(A_1 \setminus A_2) \cup (A_2 \setminus A_1), A_1 \setminus A_2, A_2 \setminus A_1 \text{ are in } \mathcal{A}_1 \text{ or } \mathcal{A}_2$$

Assume in $\mathcal{A}_1$

$\Rightarrow$ (by $\mathcal{A}_1$ is an algebra, all three sets are in $\mathcal{A}_1$)

$$\Rightarrow A_2 = \underbrace{(A_1 \cup A_2)}_{\in \mathcal{A}_1} \setminus \underbrace{(A_1 \setminus A_2)}_{\mathcal{A}_1} \in \mathcal{A}_1$$

Contradiction!

Chapter 3

Partial Differential Equations

§3.1 Lecture 1 (02-03) Initial and Boundary Conditions

§3.1.1 What are the PDEs?

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - u \quad (3.1)$$

$$u = u(x, y, t) \quad (3.2)$$

(3.1) is a second order PDE

§3.1.2 Classical solutions

To simplify, we consider here equations of one unknown function u of one or more variable

By a solution we mean a sufficiently smooth function u that satisfies the PDE at every point of the domain of its definition(D).

Sometimes, we will have to impose some boundary conditions on the domain D .

$C^n = \{u : D \rightarrow R : n\text{-times continuous differentiable w.r.t all variables} \}$ n is the order of the equation

Example 1

- Heat equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

$$u = u(x, t)$$

$$(t, x) \in D \subset R^2$$

$$u(t, x), \frac{\partial u}{\partial t}, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial t^2}$$

$$n = 2$$

$$\frac{\partial^2 u}{\partial x \partial t} = \frac{\partial^2 u}{\partial t \partial x} \in C^2$$

$$u \in C^2(D)$$

- Some Solution:

$$u(t, x) = t + \frac{x^2}{2}, D = R^2$$

$$u(t, x) = \frac{e^{-\frac{x^2}{4t}}}{2\sqrt{\pi t}}, D = \{(t, x) : t > 0\}$$

$$u(t, x) = e^{-t+ix} = e^{-t}\cos(x) + ie^{-t}\sin(x), D = \mathbb{R}^2$$

§3.1.3 Initial and Boundary Conditions

What is Initial data?

$$u(0, x) = u_0(x)$$

What are Boundary Conditions? rigorous definition of Boundary

- Dirichlet Boundary Condition: $u(t, x)|_{\partial D} = g(t, x)$
- Neumann Boundary Condition: When the normal derivative of u is specified $\frac{\partial u}{\partial n}$ where n is the normal to the boundary $\partial(D)$

$$\frac{\partial u}{\partial n}(t, x) = g(t, x)$$

- Mixed Boundary Condition: Combination of Dirichlet and Neumann

§3.1.4 Linear and Nonlinear waves

Stationary waves

$$\frac{\partial u}{\partial t} = 0 \tag{3.3}$$

$$u = u(t, x) \tag{3.4}$$

This is a first order linear homogeneous PDE

Solution of the form $u = u(t) = c, \forall c \in \mathbb{R}$

Let's integral (3.3) :

$$0 = \int_0^t \frac{\partial u}{\partial t}(s, x) ds = u(t, x) - u(0, x)$$

Thus if an initial conditional

$$u(0, x) = f(x)$$

$$u(t, x) = f(x), \forall t, x$$

Thus for any given $f \in C^1(\mathbb{R})$, we have a solution of (3.3) and this is a stationary wave

The domain of this solution is $\mathbb{R}$ if f is defined everywhere on $\mathbb{R}$

Power of domain to change the solution:

Consider the ODE:

$$\frac{du}{dt} = 0, D = (-\infty, 0) \cup (0, \infty)$$

The solution is:

$$u(t) = \begin{cases} c_1, t < 0 \\ c_2, t > 0 \end{cases} \quad \forall c_1, c_2 \in \mathbb{R}$$

similarly, for the PDE in example (3.3):

$$u(t, x) = \begin{cases} 0, x > 0 \\ x^2, x \leq 0, t > 0 \\ -x^2, x \leq 0, t < 0 \end{cases}$$

it is a $\mathbb{C}^1$ solution of (3.3) on $\mathbb{R}^2$ and it is a classical solution.

§3.2 Lecture 1 (02-05) Transport and Traveling waves

§3.2.1 uniform transport

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \quad (3.5)$$

$$u = u(0, x) = f(x) \quad (3.6)$$

$$x \in \mathbb{R}, t_0 = 0, t - t_0 = t, c \in \mathbb{R}$$

use change of variable:

$$\xi = x - ct$$

ξ is called characteristic variable

let's look for a solution of (3.5) under the form

$$u(t, x) = v(t, \xi) = v(t, x - ct)$$

use chain rule:

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} - c \frac{\partial v}{\partial \xi}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial \xi}$$

Thus:

$$0 = \frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = \frac{\partial v}{\partial t} - c \frac{\partial v}{\partial \xi} + c \frac{\partial v}{\partial \xi} = \frac{\partial v}{\partial t}$$

Conclusion:

$$\frac{\partial v}{\partial t} = 0$$

Solution is constant $v(\xi)$

$$u(t, x) = v(\xi) = v(x - ct)$$

Proposition 3.2.1

if u solves (3.5), then u has the form $u(t, x) = v(x - ct)$ where v is a function of one variable that is $\mathbb{C}^1$ -smooth.

Actually, v is satisfying

$$u(0, x) = v(x) = f(x)$$

Conclusion:

$$\begin{cases} \frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0 \\ u = u(0, x) = f(x) \end{cases}$$

The solution is

$$u(t, x) = f(x - ct)$$

$f(x - ct)$ is a traveling wave solution moving at velocity c :

if $c > 0$, the wave moves to the right

if $c < 0$, the wave moves to the left

if $c = 0$, the wave is stationary

Example 1

$$\begin{cases} \frac{\partial u}{\partial t} + 2\frac{\partial u}{\partial x} = 0 \\ u = u(0, x) = \frac{1}{1+x^2} \end{cases}$$

The solution is

$$u(t, x) = \frac{1}{1 + (x - 2t)^2}$$

The lines $x = ct + k$ are called the characteristic lines for the equation (3.5) on the characteristic lines, the solution is always constant on geometric interpretation, $u(t, x) = f(x - ct) = f(k)$

Consider the equation:

$$\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} + au = 0 \quad (3.7)$$

where a and c are given numbers.

Use change of variable:

$$u(t, x) = v(t, \xi) = v(t, x - ct)$$

we will get by the same computation

$$\frac{\partial v}{\partial t} + av = 0$$

The solution will be

$$\begin{aligned} v(\xi) &= e^{-at} f(\xi) \\ u(t, x) &= v(t, \xi) = v(x - ct) = e^{-at} f(x - ct) \end{aligned}$$

So,

$$\begin{cases} u(t, x) = e^{-at} f(x - ct) \\ u(0, x) = f(x) \end{cases}$$

§3.2.2 Nonuniform transport

$$\frac{\partial u}{\partial t} + c(x) \frac{\partial u}{\partial x} = 0 \quad (3.8)$$

In the uniform case (c constant), characteristic line is $x = ct + k$ (the solution of ODE below)

$$\frac{dx}{dt} = c$$

In the general non-uniform case, we can still define characteristic curves

$$\frac{d}{dt}x(t) = c(x(t))$$

Let's show that the solutions of (3.8) are constants along the characteristic curves $(t, x(t))$

We need to check that $u(t, x(t)) = h(t)$ is constant of t
compute the derivative of $h(t)$

$$\begin{aligned}\frac{d}{dt}h(t) &= \frac{d}{dt}u(t, x(t)) = \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} \\ &= \frac{\partial u}{\partial t} + c(x(t)) \cdot \frac{\partial u}{\partial x} = 0\end{aligned}$$

$h(t)$ and thus $u(t, x(t))$ is constant along the characteristic curves

Definition 3.2.1

The graph of a solution of the ODE

$$\frac{d}{dt}x(t) = c(x(t))$$

is called a characteristic curve for the transport equation (3.8) with speed $c(x)$

Thus at each point (t, x) the slope of the characteristic curve is $c(x)$ (the wave speed)

In the particular case $c(x)=c$, the characteristic curves are straight lines of slope c

Moreover,

(3.8) is an autonomous first order ODE which can be "solved" using the method of separation of variables

Assume that $c(x) \neq 0$

$$\begin{aligned}\frac{dx(t)}{dt} &= c(x(t)) \\ \frac{dx}{c(X)} &= dt \rightarrow \\ \beta(x) &= \int \frac{dx}{c(x)} = t + k\end{aligned}$$

Thus characteristic curves satisfies the implicit equation

$$\beta(x) = t + k$$

We can find x from the equality using β^{-1} (the inverse of β)

$$\begin{aligned}\beta(x) &= t + k \\ \beta^{-1}\beta(y) &= y \\ x(t) &= \beta^{-1}(t + k)\end{aligned}$$

Remark 0

How to determine where $c(x_0) = 0$?

Solve this ODE

$$\begin{cases} \frac{dx}{dt} = c(x) \\ x(t_0) = x_0 \end{cases}$$

and $x(t) = x_0$ will be the characteristic curve going through (t_0, x_0)

From the proposition, we know that $u(t, x(t)) = \text{constant}$

$$\begin{aligned} u(t, x(t)) &= v(k) \\ &= v(\beta(x(t)) - t) \end{aligned}$$

where k is the constant determining the characteristic curve $(t, x(t))$

$$\begin{aligned} u(t, x) &= v(\xi) \\ \text{where } \xi &= \beta(x) - t \end{aligned}$$

which we will call characteristic variable

$$u(t, x) = v(\beta(x) - t) \tag{3.9}$$

This formula may fail at points where c vanishes

$$\begin{aligned} t &= 0 \\ u(0, x) &= f(x) = v(\beta(x) - 0) \\ f(x) &= v(\beta(x)) \\ v(\xi) &= f(\beta^{-1}(\xi)) \\ &= f \cdot \beta^{-1}(\xi) \\ u(t, x) &= f \cdot \beta^{-1}(\beta(x) - t) \end{aligned}$$

This is the formula for the solution of (3.8) in the general case which is:

$$\begin{cases} \frac{\partial u}{\partial t} + c(x) \frac{\partial u}{\partial x} = 0 \\ u(0, x) = f(x) \end{cases}$$

Chapter 4

Algorithm

§4.1 Lecture 1 (02-03) Asymptotic bounds and big O

Definition 4.1.1: big-Theta notation(=)

Let $f, g : N \rightarrow R^+$, we say that $f(n) = \Theta(g(n))$ if there $\exists a, b > 0$ and $n_0 \in \mathbb{N}$ such that $a \cdot g(n) \leq f(n) \leq b \cdot g(n)$ for all $n > n_0$.

Intuitively, $f(n) = \Theta(g(n))$ means that $f(n)$ grows as fast as $g(n)$ up to a constant when n tends to infinity.

Examples: $3n = \Theta(n)$, $2n^2 + 7n = \Theta(n^2)$, $\ln n^2 = \Theta(\log n)$

Definition 4.1.2: big-O notation($\leq$)

Let $f, g : N \rightarrow R^+$, we say that $f(n) = O(g(n))$ if $\exists n_0 \in \mathbb{N}$ and $b > 0$ such that $f(n) \leq b \cdot g(n)$ for all $n > n_0$.
 $\lim_{n \rightarrow \infty} \sup \frac{f(n)}{g(n)} < +\infty$

Definition 4.1.3: big-Omega notation($\geq$)

Let $f, g : N \rightarrow R^+$, we say that $f(n) = \Omega(g(n))$ if $\exists n_0 \in \mathbb{N}$ and $a > 0$ such that $a \cdot g(n) \leq f(n)$ for all $n > n_0$.
 $\lim_{n \rightarrow \infty} \inf \frac{f(n)}{g(n)} > 0$

Remark 0

$$f(n) = \Theta(g(n)) \Leftrightarrow f(n) = O(g(n)) \text{ and } f(n) = \Omega(g(n))$$
$$f(n) = O(g(n)) \Leftrightarrow g(n) = \Omega(f(n))$$

Definition 4.1.4: little-o notation($<$)

We write $f(n) = o(g(n))$ if for any $b > 0$ there $\exists n_0 \in \mathbb{N}$ such that $f(n) < b \cdot g(n)$ for all $n > n_0$.

Definition 4.1.5: little-omega notation($>$)

We write $f(n) = \omega(g(n))$ if for any $a > 0$ there $\exists n_0 \in \mathbb{N}$ such that $a \cdot g(n) < f(n)$ for all $n > n_0$.

Remark 0

$f(n) = o(g(n)) \Rightarrow f(n) = O(g(n))$ and $f(n) \neq \Theta(g(n))$
 $f(n) = \omega(g(n)) \Rightarrow f(n) = \Omega(g(n))$ and $f(n) \neq \Theta(g(n))$
 $f(n) = \omega(g(n)) \Leftrightarrow g(n) = \omega(f(n))$

$$n^{1+e}, n \log(n), e \rightarrow 0$$

Theorem 4.1.1

Any Algorithm has to check at least $\log(n+1)$ entries of the sorted array for searching an element in the worst case

§4.2 Recitation 1 (02-07)–Problem Solving

Notation	Bounding Conditions	Relation	Type
$T(n) = O(f(n))$	$\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \leq cf(n)$	$\leq$	O
$T(n) = \Omega(f(n))$	$\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \geq cf(n)$	$\geq$	Ω
$T(n) = \Theta(f(n))$	$\exists c_1, c_2 > 0, \exists n_0, \forall n \geq n_0, c_1 f(n) \leq T(n) \leq c_2 f(n)$	$=$	Θ
$T(n) = o(f(n))$	$\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) < cf(n)$	$<$	o
$T(n) = \omega(f(n))$	$\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) > cf(n)$	$>$	ω

Possible useful formula:

$$\int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$

1. Multiplication Rule. Prove that for $h(n) = \Omega(1)$, if $f(n) = \Theta(g(n))$ then $f(n) \cdot h(n) = \Theta(g(n) \cdot h(n))$ (The same also holds for $O, \Omega, \omega, \omega$)

Solution:

$$\exists c_1, c_2 > 0, \exists n_0, \forall n \geq n_0, c_1 g(n) \leq f(n) \leq c_2 g(n)$$

$$\exists c_3, c_4 > 0, \exists n_1, \forall n \geq n_1, c_3 h(n) \leq f(n) \leq c_4 h(n)$$

$$\text{Let } c_5 = c_1 c_3, c_6 = c_2 c_4$$

$$\text{Let } n_2 = \max(n_0, n_1)$$

$$\forall n \geq n_2, c_5 g(n) h(n) \leq f(n) h(n) \leq c_6 g(n) h(n)$$

$$\therefore f(n) h(n) = \Theta(g(n) h(n))$$

2. Addition Rule. Let $f_1, \dots, f_k : N \rightarrow R_{\geq 0}$ for a fixed number k and $f(n) = \sum_{i=1}^k f_i(n)$. Prove that if $f_i(n) = O(f_1(n))$ for all $i \in \{1, \dots, k\}$, then $f(n) = \Theta(f_1(n))$. Here

f_1 can be replaced with any one of $f_1, \dots, f_k$

Solution:

$$\exists c_{1i} > 0, \exists n_{0i}, \forall n \geq n_{0i}, f_i(n) \leq c_{1i} f_1(n)$$

$$\text{Let } c_1 = \sum_{i=1}^k c_{1i}$$

$$\text{Let } n_0 = \max(n_{01}, \dots, n_{0k})$$

$$\forall n \geq n_0, \sum_{i=1}^k f_i(n) \leq \sum_{i=1}^k c_{1i} f_1(n)$$

$$\therefore f(n) = \Theta(f_1(n))$$

3. Let f and g be non-negative functions. Define $T(n) = \max(f(n), g(n))$. Prove that $T(n) = \Theta(f(n) + g(n))$

Solution:

$$\forall n \geq n_0, f(n) \leq T(n) \leq f(n) + g(n)$$

$$\forall n \geq n_0, g(n) \leq T(n) \leq f(n) + g(n)$$

$$\therefore T(n) = \Theta(f(n) + g(n))$$

4. Let f and g be non-decreasing real-valued functions defined on positive integers and $f(1) \geq 1$ and $g(1) \geq 1$. Prove or disprove that $f(n) = O(g(n)) \Rightarrow 2^{f(n)} = O(2^{g(n)})$

Solution:

$$\exists c > 0, \exists n_0, \forall n \geq n_0, f(n) \leq cg(n)$$

$$\forall n \geq n_0, 2^{f(n)} \leq 2^{cg(n)}$$

$$\text{Let } c_1 = 2^c$$

$$\forall n \geq n_0, 2^{f(n)} \leq c_1 \cdot 2^{g(n)}$$

$$\therefore 2^{f(n)} = O(2^{g(n)})$$

5. Prove that $\log(n!) = \Theta(n \log n)$. Hint: To show an upper bound, compare $n!$ with n^n . To show a lower bound, compare it with $\left(\frac{n}{2}\right)^{\frac{n}{2}}$

Solution:

$$n! = n \cdot (n-1) \cdot \dots \cdot 2 \cdot 1 \leq n \cdot n \cdot \dots \cdot n \cdot n = n^n$$

$$n! = n \cdot (n-1) \cdot \dots \cdot \left(\frac{n}{2} + 1\right) \cdot \left(\frac{n}{2}\right) \cdot \dots \cdot 2 \cdot 1 \geq \left(\frac{n}{2}\right)^{\frac{n}{2}}$$

$$\therefore \log(n!) = \Theta(n \log n)$$

6. Show that $c^n = o(n!)$ for any constant C

Solution:

$$\forall n \geq n_0, c^n < n!$$

$$\text{Let } n_0 = 2$$

$$\forall n \geq n_0, c^n < n!$$

$$\therefore c^n = o(n!)$$

7. Show that $c^{n^{1.001}} = \omega(n!)$ for any constant C . Hint: For any fixed numbers $p, q > 0$, we have $n^p = \omega(\log^q n)$

Solution:

$$\forall n \geq n_0, c^{n^{1.001}} > n!$$

$$\text{Let } n_0 = 2$$

$$\forall n \geq n_0, c^{n^{1.001}} > n!$$

$$\therefore c^{n^{1.001}} = \omega(n!)$$

8. Show $\sum_{i=1}^n i^k = \Theta(n^{k+1})$. Hint: Show that $n^{k+1} - (n-1)^{k+1} = \Theta(n^k)$

Solution:

$$n^{k+1} - (n-1)^{k+1} = \Theta(n^k)$$

$$\sum_{i=1}^n i^k = \Theta(n^{k+1})$$

9. Search for the peak. Suppose we have an array $A[1 \dots n]$ and we know that there exists some k (value unknown) such that $A[1] < A[2] < \dots < A[k]$ and $A[k] > A[k+1] > \dots > A[n]$. We call $A[k]$ the peak of A . Design an algorithm to find the peak and analyze it in $O(\log n)$ time.

Solution:

Algorithm:

1. Set $l = 1, r = n$

2. While $l < r$ do

3. Set $m = \left\lfloor \frac{l+r}{2} \right\rfloor$

4. If $A[m] < A[m+1]$ then $l = m+1$

5. Else $r = m$

6. Return l

Analysis:

Let $T(n)$ be the time complexity of the algorithm

Let $n_0 = 1$

Let $c = 1$

Let $T(n) = T\left(\frac{n}{2}\right) + c$

Let $T(n) = O(\log n)$

10. 3SUM in a sorted array. Let $A[1 \dots n]$ be a sorted array. We want to see $A[i] + A[j] + A[k] = x$ for some different i, j, k , where x is a given number. Design an algorithm $3SUM(n, a, x)$ which can return (i, j, k) or 0 if no such three elements exist. How fast can this problem be solved?

Solution:

Algorithm:

1. For $i = 1$ to $n - 2$ do
2. Set $l = i + 1, r = n$
3. While $l < r$ do
4. Set $s = A[i] + A[l] + A[r]$
5. If $s = x$ then return (i, l, r)
6. Else if $s < x$ then $l = l + 1$
7. Else $r = r - 1$
8. Return 0

Analysis:

Let $T(n)$ be the time complexity of the algorithm

Let $n_0 = 1$

Let $c = 1$

Let $T(n) = T(n - 1) + c$

Let $T(n) = O(n)$